Extended Data Fig. 10 | RNA velocity consistency with and without non-target spots

Paired direction consistency for the same 608 target spots in the 1,292-spot ST_raw dataset and the 608-spot ST_small subset. a, Per-spot cosine similarity of gene-space velocities on each method's shared finite genes; the number of genes is shown at right. b, Cosine similarity of spatial directions, with medians at right. Boxes show the median and interquartile range; whiskers show the 5th and 95th percentiles. c, Median spatial cosine for Inner EPC1 (47 spots), Inner EPC2 (87), Outer EPC1 (64), Outer EPC2 (61), Secondary P-TGC1 (153), and Secondary P-TGC2 (196). d, Per-spot spatial cosines without spatial smoothing; the other 684 spots are grey, and medians appear in parentheses. Red denotes positive cosine (same direction at +1), white denotes zero (perpendicular), and blue denotes negative cosine (opposite at -1) in c-d. All panels use the descending spatial-median order. Each of 19 methods has 608 valid spatial pairs. Gene comparisons have 608 valid pairs for 17 conventional gene-velocity outputs and the TopicVelo proxy; VeloAE is unavailable because its independently learned latent axes cannot be matched. TopicVelo uses the previously rerun native transitions: its gene-space quantity is a transition-displacement proxy T @ Ms - Ms evaluated in one fixed full-data expression reference, and its spatial directions use the native transition matrices. Eleven and five usable topics are integrated for ST_raw and ST_small, respectively. Spot IDs and annotations were matched; a constant horizontal coordinate translation of 150 units was verified and does not affect directions. Shared gene sets differ across methods (39-1,259 genes). Preprocessing, retained genes, neighborhoods and fitted models differ between the two saved runs. This is a descriptive comparison of saved-output sensitivity, not a controlled estimate of the isolated effect of non-target spots. Spots are observations from one supplied dataset, not independent biological replicates. No new model fitting or significance testing was performed; consistency does not establish directional accuracy or velocity magnitude.